

ZHOU XUYAN

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EDUCATION

National University of Singapore (NUS) Bachelor of Computing in Computer Science Second Major in Mathematics Award: NUS Science and Technology Scholarship	<i>Aug 2022 – Jul 2026</i> <i>Singapore</i>
Temasek Junior College (TJC) SM1 Scholarship A'Level 88.75/90	<i>Jan 2016 - Dec 2019</i>

EXPERIENCE

School of Design and Environment <i>Research Assistant</i>	<i>Mar 2026 – Jun 2026</i> <i>Singapore</i>
- Partnered with DfS domain experts to translate construction-safety requirements into an end-to-end multimodal recommendation system using Qwen VLM, ChromaDB RAG, and Neo4j KG-RAG. - Engineered ingestion pipelines to extract, normalise, validate, and load 1,000+ DfS case studies, registers, and regulatory documents into structured schemas, vector stores, and a knowledge graph. - Benchmarked Qwen-only, Qwen+BGE, and Qwen+BM25 retrieval pipelines to optimise recommendation quality, building structured evaluation workflows for human validation.	
ByteDance <i>Seed - LLM Global Data Operations Intern</i>	<i>Aug 2025 - Nov 2025</i> <i>Singapore</i>
- Evaluated 5+ Seed-developed LLMs on coding tasks and applied supervised CoT fine-tuning, improving performance on the SWE benchmark by 6%. - Built automated web crawlers and data pipelines for ICPC/CCPC and Codeforces, replacing external data vendors and reducing associated data acquisition costs by 100%. - Translated R&D training data requirements into standardised processing guidelines and quality controls, coordinating with external vendors to ensure consistent delivery for LLM training pipelines.	
Singtel <i>Management Associate Program Intern</i>	<i>May 2025 - Aug 2025</i> <i>Singapore</i>
- Built and operated ELK/Elasticsearch dashboards for real-time application and error monitoring, reducing debugging time by 50% and operational costs by 66%. - Partnered with internal security and business stakeholders to translate training requirements into a gamified AI-security platform focused on deepfake and scam awareness. - Improved reliability of HiApp core services by designing 200+ JUnit test cases across functional and edge-case scenarios, increasing test coverage from 39% to 89%.	
ST Engineering <i>Software Engineering Intern</i>	<i>May 2024 - Oct 2024</i> <i>Singapore</i>
- Built a reusable OutSystems application template adopted across 10+ projects, standardising common functionality and accelerating delivery across multiple teams. - Developed role-based access control and delivered enhancements and production fixes across 5+ applications, supporting secure and reliable releases. - Led UAT for 3 applications, translating end-user feedback into pre-deployment fixes and creating 5+ technical guides/scripts to support rollout and adoption.	

SELECTED PROJECT

zkVM Architecture & Performance Benchmarking — Undergraduate Thesis
Implemented stack- and register-based zkVM interpreters and built a Python benchmarking framework across 7 workload families, identifying that register-based execution reduced total constraints by 20–29% at larger workloads.

SKILLS

Programming: Python, Java, SQL
AI/ML: LLMs, VLMs, RAG, Agentic AI, NLP, Supervised Fine-Tuning, Retrieval & Reranking
Data & Systems: ChromaDB, Neo4j, Elasticsearch/ELK, Data Pipelines, Web Crawling
Developer Tools: Codex, Claude Code, JUnit, OutSystems
Languages: English, Chinese (Native)